AI Methods for Classification of Production Errors on Camera Images of Additive Manufacturing Processes

universität koblenz
weiter:denken

Ramya Subramanian Porselva Bharathi
Matriculation Number: 224201309

The Problem

Additive Manufacturing (AM) builds parts layer-by-layer, used in aerospace, automotive & biomedical - but prone to defects: porosity, lack of fusion, surface roughness, and delamination.

Defects compromise structural integrity and increase production cost and time.

Camera-based monitoring is non-contact and low-cost, but rule-based methods achieve only 65–75% accuracy due to lighting variability and environmental factors.

→ **AI-based automation is needed.**

Research Scope

Systematic review of AI methods for camera-based defect classification in AM, covering publications 2018–2025.

Three method families reviewed:

1. Traditional Image Processing
2. Classical ML (SVM, Random Forest)
3. Deep Learning (CNN, YOLO, U-Net)

Method Comparison

Approach	Typical Accuracy	Relative Speed	Key Limitations
Traditional Image Processing	Low–moderate accuracy	Very fast (>100 fps)	Illumination sensitivity, manual recalibration
Classical ML (SVM, RF)	Moderate accuracy	Moderate (20-50 fps)	Feature engineering, limited generalization
Deep Learning (CNN)	High accuracy	Moderate	Data requirements, interpretability challenges
Object Detection (YOLO)	High localization accuracy	Real-time capable (11-15 fps)	Computational cost, domain shift

Key Results, Challenges & Future Directions

Key Findings

1. Deep learning consistently outperforms traditional and classical methods - CNNs & YOLO-based models achieve 92–97% accuracy, outperforming traditional methods by 20–30 percentage points.
2. YOLO-based models are the most promising for industrial deployment - enabling real-time simultaneous defect localization & classification (>90% mAP, ~15 fps).
3. Transfer learning reduces labeled data requirements by factors of 10–50×, enabling strong performance even with limited domain-specific datasets.

Challenges Limiting Industrial Deployment

Dataset scarcity: Few large-scale public datasets; annotation requires expert knowledge
Domain shift: Models degrade when hardware, materials, or lighting conditions differ
Edge deployment: Real-time inference on embedded/edge platforms remains difficult
Interpretability: Black-box models hinder regulatory acceptance in safety-critical sectors

Future Directions

Short-term: Standardized benchmark datasets & lightweight edge models
Mid-term: Closed-loop process control integration & multi-modal sensor fusion
Long-term: Physics-informed neural networks & fully autonomous quality assurance